

Skateboard Simulator

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1. Project Overview

The skateboard simulator was developed using Unreal Engine 5.5.4. The player's movement system is based on the ACharacter class of Unreal Engine. Initially, the implementation of a custom movement and physics system using the APawn class was considered, but due to time constraints, the standard Unreal movement system was adopted.

2. Character and Movement Implementation

Development began with a Third Person Template, where the ASkateCharacter class was created. The first steps involved setting up the camera and the character's SkeletalMesh. Subsequently, the focus shifted to skateboard movement:

- **Forward Movement:** The player moves forward by pressing the "W" key, with the direction based on the SkeletalMesh's forward vector.
- **Lateral Movement:** By pressing the "A" or "D" keys, the character's mesh rotates in the desired direction, allowing for lateral movement.

3. Speed Control

Speed control was implemented by adjusting the character's maximum speed (Character Max Speed). This allows the player to completely stop the character by releasing the "W" key. To increase the maximum speed, the player can hold down the "Shift" key. Similarly, to decrease speed, the "S" key is used, reducing the maximum speed.

Minimum and maximum values were defined for the maximum speed to prevent the player from reaching unnecessarily high speeds or from becoming static or having negative speed.

4. Obstacles and Scoring System

Obstacles are based on the AActor class and feature a Static Mesh and a raycast system for score calculation. The scoring system adds one point each time the player jumps over an obstacle. This score is granted only once per obstacle, a decision made to simplify coding and provide a more responsive point system.

5. Map Design

The map was designed to guide the player towards obstacles and teach them that jumping over them grants points. Additionally, the map design offers diversity in obstacle locations, aiming for a varied gameplay experience.

6. User Interface (HUD)

The user interface (HUD) was kept simple, displaying basic information to the player, such as points and speed. It also shows which keys the player can use throughout the playthrough, making it easier to understand the controls.